

DIGITAL ASSETS
OCTOBER 2025

An Investor's Guide to Bitcoin & Altcoin Treasury Companies

Premiums, Yields, and Structural Risks

Martin Leinweber, CFA
Director Digital Asset Research & Strategy

 MarketVector™

Contents

Executive Summary	2
Investment Highlights	2
Bottom Line for Investors	3
1. What Makes a Bitcoin Treasury Company Different from an ETF	3
2. Why Premiums to NAV Exist (and Why They're Fragile)	4
3. How "Strategy's" Capital Structure Works (High Level)	5
4. Can Treasury Companies Generate Income/Yield Without Selling Bitcoin?	5
5. The "Ponzi-Adjacent" Problem (What It Is—and Isn't)	6
The Investment Framework	6
Positioning Takeaway	8
DATs 2.0: Treasury Companies on Altcoins	8
How Altcoin DATs Differ from BTC Treasury Companies	8
Pros of Altcoin DATs	8
Cons / Risks of Altcoin DATs	9
Analogies to BDCs and Listed PE: Lessons on Premium Sustainability	10
Takeaway	10
Key Terms Explainer	11

Executive Summary

DAT companies are publicly listed corporations that accumulate Bitcoin or Altcoins as core assets. Unlike ETFs, they finance growth through equity, convertibles, and preferred stock, making them **structured**, **cycle-sensitive exposures** rather than simple token proxies.

Investment Highlights

Not ETFs — Structured Exposures

DATs introduce **capital structure dynamics**: premiums/discounts to NAV, financing strategies, and refinancing risk. Outperformance in bull markets stems from **accretive issuance** and, for altcoins, **staking/ DeFi yields**.

Premiums: The Core Driver

- **High MNAV (>1)**: Enables accretive issuance and compounds BTC/token per share.
- **Fragility**: Premiums often compress in bear phases, when liquidity tightens, or if index eligibility changes.

Capital Structures & Risk Factors

- **Financing tools**: converts, preferreds, and at-the-market (ATM) equity programs.
- **Risks**: stalled issuance, dividend accrual, refinancing stress, and—if liquidity runs short—forced token sales.

Yield Dynamics

- **Bitcoin DATs**: Limited recurring yield; rely on capital-market mechanics.
- **Altcoin DATs**: Staking yields (5–8% APY) can enhance NAV, but introduce validator, smart-contract, and lock-up risks.

Investor Framework

- View BTC DATs as levered beta to digital assets, not as cash-flow businesses.
- Key indicators: MNAV trend, token per share growth, preferred terms, liquidity runway.
- Best entry points: moderate premiums, disciplined issuance, long dated/non-callable liabilities.

Regulatory & Index Considerations

- SEC classification risk for SPAC/RTO structures.
- Index exclusions may limit liquidity and valuations.
- Likely outcome: only a few leading DATs per major token survive long term.

Bottom Line for Investors

DATs offer **levered access to Bitcoin and Altcoins**, with potential outperformance in bull cycles. But they also embed **fragile capital structures** that can underperform sharply in downturns. Treat them as **tactical portfolio tools**, sized appropriately within a broader digital asset allocation.

Introduction

This old maxim applies well to the emerging category of crypto treasury companies—publicly traded corporations that accumulate Bitcoin or other coins as a core asset through strategic financing, often emulating models like MicroStrategy's. Many investors approach these equities thinking they are simple coin proxies, only to discover that they are complex corporate structures with financing mechanics, capital market dependencies, and cycle-sensitive risks.

“If you’ve been playing poker for half an hour and you still don’t know who the patsy is, you’re the patsy.”
Warren Buffett

This primer begins with a focused analysis of Bitcoin treasury companies—their mechanics, opportunities, and risks—and concludes with a comparison to the latest developments in Digital Asset Treasury (DAT) companies emerging around Ethereum, Solana, and other altcoins. The purpose is not to praise or criticize the model, but to explain it.

Investors need to know what they are buying: how these companies can outperform in bull markets, how they can underperform or stress in bear markets, and how to monitor the signals that matter. Understanding this structure is essential for two conversations at once—with the companies themselves, and with clients considering investment.

1. What Makes a Bitcoin Treasury Company Different from an ETF

Objective & Mandate

- ETF (spot): Holds BTC 1:1, aims to track NAV via primary/secondary market arbitrage. No discretionary leverage, no “strategy,” no management of a balance sheet.
- Treasury company (“Strategy” archetype): A corporation that actively finances BTC accumulation using equity, convertibles, and preferreds. It can increase BTC per share by selling over-NAV equity or issuing liabilities to buy more BTC. Example: MicroStrategy (MSTR) has grown its BTC holdings to ~ 630,000 as of mid-2025 through such issuances.

Where “Alpha” Can Come From

- ETF: None by design—pure beta.
- Treasury company: Potential financing alpha (term leverage + accretive issuance when trading at a premium) and management/timing of the capital stack. **This is not return on BTC; it is return from capital markets mechanics.**

Risk Profile

- ETF: Tracks BTC; structure risk is minimal and transparent.
- Treasury company: Adds capital-market and refinancing risk (premiums can compress; access to issuance can shut; coupons/dividends may be paused).

2. Why Premiums to NAV Exist (and Why They're Fragile)

Premiums arise as the market prices in perceived value beyond the underlying crypto holdings, reflecting structural edges and sentiment-driven dynamics over simpler proxies like ETFs. However, they are inherently fragile, often evaporating in bear phases or with increased competition. Key drivers include:

Accretive Issuance Flywheel:

When shares trade above NAV ($MNAV > 1$), the company can sell \$X of stock, buy ~\$X of BTC, and—because it sold at a premium—raise BTC/share for continuing holders. The higher the premium, the faster BTC/share can grow.

This is recursive:

A high premium enables more issuance, which boosts BTC/share growth, justifying the premium—until sentiment or liquidity shifts. Hypothetical: At MNAV 1.5x, issuing \$100M buys \$100M BTC but accretes ~1.5% to BTC/share, as you issue fewer new shares than if you issued at NAV parity.

Financing Access & Term Leverage:

Corporates can attach long-dated, non-callable liabilities (e.g., converts, preferreds) to BTC—something a spot ETF doesn't do. Investors may pay a premium for that packaging, as it amplifies beta without immediate cash drag.

Perceived Management Edge & Liquidity:

Some investors want a levered, high-beta BTC proxy with deep single-name liquidity and management they trust to operate the balance sheet. For instance, strong leadership (e.g., MicroStrategy's under Michael Saylor) adds a "convenience premium" for operational expertise and collateral usability.

Path Dependence & Regime Shifts:

Historically, lack of easy BTC access (pre-ETF) supported big premiums. Even post-ETF, premiums can persist in bull phases—but tend to compress as markets mature or turn risk-off.

Market Sentiment and Narrative Momentum:

In expansions, premiums embed hype around compounding potential, meme-like virality, and crypto's scarcity narrative (e.g., BTC's 21M cap), driving FOMO. This is the most fragile driver, as seen in analogs like Grayscale's GBTC premium collapse post-ETF launches.

3. How “Strategy’s” Capital Structure Works (High Level)

Legacy converts (low/zero coupon):

Earlier cycles favored converts with minimal near-term cash costs; refinancing risk sits at maturity.

Shift to preferreds:

Recent issuance emphasizes preferred equity with stated dividends. Two flavors matter:

- Cumulative senior preferreds: If dividends are paused, they accrue and must be caught up later before common can resume payouts.
- Junior/non-cumulative preferreds: Dividends can be paused without accrual, effectively shifting stress to preferred holders.

ATM equity programs:

When MNAV is high, the company issues common stock opportunistically to buy more BTC (accretive to BTC/share).

Failure path the market worries about:

BTC drawdown or long sideways period → MNAV compresses → equity issuance window narrows → preferred obligations loom → dividends paused (junior first) → preferreds reprice hard → common reprices below NAV (to reflect accrued obligations) → potential forced BTC sales if stress persists.

4. Can Treasury Companies Generate Income/Yield Without Selling Bitcoin?

Today: Mostly no, not at scale. Mark-to-market gains (now GAAP-visible under FASB ASU 2023-08) are not recurring operating income and shouldn’t command a conventional earnings multiple.

What exists but is small/early:

- Payments liquidity (e.g., Lightning): Some firms report BTC-denominated ROI from routing/liquidity provision, but scale and uniformity are limited; numbers vary widely and are not enterprise-sized yet.
- BTC lending/covered options: Possible in theory, but introduces counterparty, rehypothecation, or capped-upside risks; many investors deem these unacceptable post-FTX in corporate treasuries.

Bottom line:

For now, the “yield” engine is capital markets, not the Bitcoin stack. Income that reliably services coupons/dividends without new issuance is the missing piece.

5. The “Ponzi-Adjacent” Problem (What It Is—and Isn’t)

What critics mean:

If preferred dividends (or other cash obligations) are funded primarily by new security issuance—because the asset (BTC) doesn’t produce cashflows—then returns to existing holders depend on new capital. That is Ponzi-adjacent even if disclosed and entirely legal.

When it can unravel:

1. MNAV compression toward/below 1 removes accretive issuance.
2. Capital market access tightens (risk-off, regulatory shock, issuer fatigue).
3. Prolonged BTC stagnation (not just a sharp drawdown) starves the model of windows to refinance/issue.
4. Dividend pauses (especially on cumulative preferreds) create stacked claims that subordinate the common and depress both preferred and common valuations.
5. If stress persists, forced BTC sales can create a negative feedback loop (lower asset base → weaker coverage → worse terms).

Why it’s not outright fraud:

The mechanics are transparent corporate finance, not hidden books. But marketing mark-to-market gains as “recurring earnings that deserve a multiple” is misleading—they are volatile asset revaluations, not durable cashflows.

The Investment Framework

Why Own a Well-Run Treasury Company at All?

- You want levered BTC beta with professional access to term financing and the chance to compound BTC/share via accretive issuance in bull regimes.
- You accept capital-structure risk in exchange for potential outperformance vs. spot BTC during expansion phases.

When It Can Make Sense

- Moderate leverage, long-dated/non-callable liabilities, and clear stress-test thresholds for bear markets (including explicit dividend-pause playbooks).
- Disciplined issuance (sell when MNAV is rich; slow/stop when it’s tight), and transparent dashboards for BTC/share growth, MNAV bands, and runway.
- Entry when MNAV is reasonable rather than euphoric.

What to Monitor (Early Warning Set)

- **MNAV level & trend:** Sustained compression = issuance risk.
- **BTC/share trajectory:** Stalling or dilution without offsetting BTC adds is a red flag.
- **Preferred mix & terms:** Cumulative vs. non-cumulative, step-ups, and deferral mechanics; accrual build-up.
- **Debt maturities & covenants:** Refunding risk profile through a full cycle.
- **Capital market windows:** Actual volumes in ATM sales and pricing of new paper.
- **Management messaging:** Any earnings-multiple talk on mark-to-market gains deserves skepticism.

MNAV < 1: Do They Have to Sell BTC?

Short answer: No. Trading below NAV doesn't by itself force sales.

- **General Mechanics:** Issuance stalls (new common becomes dilutive to BTC/share, so management typically slows/halts ATM issuance). Use “pressure valves”: Pause or defer preferred dividends (per terms), cut costs, and seek refinancing before touching BTC. No automatic margin call unless there's secured, BTC-collateral debt with LTV triggers.
- **When Sales Can Happen:** Only if there's a real liquidity shortfall (opex + interest + required dividends) and no accretive equity window or refinancing remains, after using deferral/pausing levers. MSTR-Specific Note: With no active BTC-collateral loan, MNAV < 1 alone doesn't mandate sales. The likely playbook: Pause issuance → conserve cash → pause certain preferreds per terms → consider asset sales only if runway is exhausted.

Is It Value-Accretive to Sell BTC and Buy Back Stock When MNAV < 1?

Often yes—mathematically—but it's a policy and constraints question.

- **Quick Tests:**
 - **NAV/share accretion:** Buyback price < NAV/share ⇒ NAV/share rises for remaining holders.
 - **BTC/share accretion (rule of thumb):** If Market Cap < BTC Asset Value, using \$1 of BTC to retire \$1 of equity retires a larger % of shares than % of BTC sold ⇒ BTC/share rises. (MNAV < 1 implies EV < BTC value, so Market Cap < BTC value is typically true.
- **Why They Still Might Not Do It:** Mandate/strategy (accumulate BTC, not sell); taxes & frictions (gains on sales); covenants/terms (restrictions if dividends paused); runway & cushion (reduces asset buffer); signaling (undercuts “never sell” narrative).
- **Bottom line:** When equity trades at a meaningful discount, BTC-funded buybacks are usually accretive to both NAV/share and BTC/share. Whether management should do it depends on taxes, covenants, liquidity runway, and stated strategy.

Positioning Takeaway

Treat these equities as structured, cycle-sensitive BTC exposures—not cash-flow businesses. The upside is real in bull phases; the downside appears when the model becomes dependent on issuance that the market won't fund. Hold spot BTC (or an ETF) as your core and size treasury-company exposure tactically, keyed to MNAV and liquidity conditions. For diversification, consider blending with altcoin DATs for yield enhancement, but manage positions according to your capacity of taking risks.

DATs 2.0: Treasury Companies on Altcoins

Building on the Bitcoin treasury model, a new wave of DAT companies are emerging around altcoins such as Solana (SOL) and Ethereum (ETH) and other altcoins. These entities borrow the playbook—raising equity and convertibles, buying tokens, aiming to compound NAV/share—but differ in economics, risks, and opportunities. Examples: Bitmine Immersion (BNMR) for ETH-focused treasuries; early Solana DATs like those from SOL Treasuries and Forward Industries (FORD), the latter led by Kyle Samani of Multicoine Capital with a \$1.65B PIPE in September 2025 to build SOL holdings and on-chain infrastructure.

How Altcoin DATs Differ from BTC Treasury Companies

1. Underlying Asset Characteristics

- **Bitcoin DATs:** BTC is non-yielding, liquid, and viewed as digital gold. Value accrual comes from capital-market mechanics.
- **Altcoin DATs:** Tokens like SOL or ETH generate yield through staking/restaking. DATs can buy locked tokens at discounts, enhancing returns. Example: SOL staking yields ~5-8% APY in 2025.

2. Financing and Accretion Model

- Bitcoin DATs: Reliant on MNAV premiums for accretion; fragile if compressed.
- Altcoin DATs: Same flywheel, but token yields compound NAV faster. Equity above book buys yield-bearing assets.

3. Investor Education and Branding

- Bitcoin DATs: Established (e.g., MSTR leadership).
- Altcoin DATs: Heavier education needed; success requires highlighting yield edges over BTC.

Pros of Altcoin DATs

- **Yield Accrual:** Staking turns treasuries productive, boosting returns.
- **Higher Growth Runway:** Smaller caps (SOL ~4% of BTC) embed upside.
- **Diversification:** Broader crypto exposure if invested in a basket of token.
- **Potential for Meme/Attention Multipliers:** Fundamentals plus virality accelerate premiums.
- **Institutional Backing Potential:** High-profile launches (e.g., Ford's \$ 1.65B PIPE) can attract sticky capital providing initial escape velocity¹.

¹ <https://www.businesswire.com/news/home/20251006680092/en/Forward-Industries-Announces-Launch-of-Solana-Validator-and-Integration-with-DoubleZero>

Cons / Risks of Altcoin DATs

- **Premium Fragility and Saturation:** Many entrants compress MNAV.
- **Operational Risks: Staking/ DeFi** exposes to slashing, smart-contract failures; lock-ups illiquid in stress.
- **Governance and Discretion:** Management choices in validators/issuance critical.
- Regulatory and Index Inclusion Risks
- **SEC Scrutiny on Formation and Classification:** Even under crypto-friendly SEC Chair Paul Atkins, DATs formed via SPACs or reverse takeovers (RTOs) of impaired shells face hurdles. SPACs must merge with "operating businesses" per rules; acquiring passive asset-holders risks reclassification as Investment Companies under the 1940 Act, potentially blocking deSPAC approvals². RTOs of non-crypto firms (e.g., pharma shells) often abandon legacy operations, raising questions on whether staking alone qualifies as "operating"—unlikely sufficient for Bitcoin PoW DATs, which lack yields without riskier DeFi/lending. Trailblazers like MSTR got a pass via organic evolution; new DATs may need proactive business builds or acquisitions.
- **Index Exclusion and Liquidity Impacts:** FTSE Russell's ground rules explicitly govern SPAC eligibility: unmerged SPACs or entities that remain "non-operating structures" must convert into an operating company or file a qualifying IPO registration before being considered for inclusion³. DATs that lack genuine operations risk being deemed ineligible for Russell 1000/2000/3000, potentially missing a large tranche of passive buying (often cited in index flows). Many 2025 DATs structured via sub-eligible shells may not satisfy the SPAC-to-operating-company transition before quarterly reviews, and thus may struggle to be added during IPO windows or at June reconstitution—raising the risk of discount-to-NAV (sub-MNAV) trading and an impaired flywheel effect. Historically, in 2014, MSCI, Russell and S&P removed numerous BDCs under the AFFE regime (which required funds to include acquired-fund expenses), precipitating valuation and volume declines⁴. That said, treasury-holding companies (e.g. firms using crypto treasuries) may still qualify for inclusion in **digital-asset thematic indexes** such as the [MVIS Global Digital Assets Equity Index](#), depending on respective index inclusion rules.
- **Attention Scarcity:** Only top 2-3 DATs per token likely survive.

² <https://www.sec.gov/files/rules/final/2024/33-11265.pdf>

³ <https://research.ftserussell.com/products/index-notices/home/getmethodology/?id=2615127>

⁴ <https://sbia.org/2025/03/18/bipartisan-bill-would-remove-obstacle-to-bdc-investment/>

Analogies to BDCs and Listed PE: Lessons on Premium Sustainability

To frame expectations for MNAV premiums, one can draw analogies to Business Development Companies (BDCs) and listed private equity (PE) vehicles, both of which exhibit NAV-driven valuation dynamics—yet also expose vulnerabilities relevant for DATs.

BDCs, regulated under the 1940 Act, invest in private debt/equity and trade at premiums/discounts based on yields and cycles—often compressing to 10-15% discounts in stress, as seen in Q2 2025 amid rising rates. Their 2014 Russell exclusion (due to fund-like classifications) led to liquidity drops and persistent sub-NAV trading, a cautionary parallel for DATs lacking operations⁵.

Listed PE firms, such as closed-end private equity trusts, have historically traded at persistent discounts—often ~10–20 % (sometimes deeper in stressed environments)—despite having underlying value in illiquid holdings⁶. Their discounts widen during periods of uncertainty, reinforcing that even high-brand names with fee models are vulnerable to NAV dislocation.

Thus, for DATs that lack BDC-style yield engines or durable fee streams, analogies suggest potential evolution toward lower premiums or extended discount phases (e.g. 10–30 %+). A prudent approach would be to enter at mild premiums, maintain risk awareness around index inclusion, reconstitution events, and discount catalysts, and monitor for structural discount triggers.

Aspect	DATs	BDCs	Listed PE
NAV Premium/Discount Drivers	Issuance flywheel, crypto sentiment	Portfolio yields, credit cycles	Fee income, fund performance
Avg. 2025 Discount	0-30% in stress (emerging)	10-15% (widening Q2)	20-40% (historical avg.)
Income Source	Staking (altcoins); none (BTC)	Loan interest/fees	Management fees/carry
Regulatory Risk	High (1940 Act potential)	High (AFFE/index exclusion)	Moderate

Takeaway

Altcoin DATs extend the BTC model into yield-bearing assets, offering greater upside but more risks. Altcoin DATs extend the BTC model into yield-bearing assets, offering greater upside but more risks. To sustain, they must evolve beyond asset accumulation—building or acquiring operating businesses early to navigate SEC/Russell gates and avoid "perma-purgatory." Winners will be few: 2–3 per major token with strong policy, issuance, and brand. They complement Bitcoin treasuries as high-beta proxies.

Treasury companies offer structured access to digital assets, but require vigilant monitoring of cycle dynamics. As altcoin DATs mature, they may diversify the category—yet the core lesson remains: Understand the flywheel, or risk being the patsy.

⁵ <https://alterdomus.com/insight/bdcs-a-menu-of-risk-reward-options/>

⁶ "Net asset value discounts in listed private equity funds" (Authors: Kaserer, Christoph; Lahr, Henry)

Key Terms Explainer

NAV (Net Asset Value)

Value of BTC holdings **minus** liabilities attributable to those assets (if you keep it simple, use market value of BTC minus net debt) **per share**. ETFs are designed to trade ~at NAV.

Enterprise Value (EV)

Market cap + total debt – cash. Use EV when comparing the market's value of the business to the value of its BTC.

MNAV (Market-to-NAV multiple)

How rich/cheap the equity is vs. its BTC stack.

Definition (practical): $MNAV = EV \div \text{fair value of BTC holdings}$.

- **MNAV > 1:** trades at a **premium** (market paying for leverage/issuance ability/management).
- **MNAV < 1:** trades at a **discount** (flywheel pressure; issuance becomes harder). Premium/discount $\approx MNAV - 1$.

BTC/share

Bitcoin exposure **per common share**: $\text{BTC holdings} \div \text{fully diluted share count}$. Key north star—rising BTC/share = compounding exposure.

Accretive vs. dilutive issuance

Issuing new common above NAV (high MNAV) and using proceeds to buy BTC **raises BTC/share** (accretive). Issuing near/below NAV **lowers BTC/share** (dilutive).

ATM (At-the-Market) program

Drip-selling new shares directly into the market. Powerful when MNAV is high; typically paused when MNAV ~1 or below.

Preferred stock (prefs)

Sits **senior to common, junior to debt**. Usually pays a fixed dividend.

- **Cumulative:** skipped dividends **accrue**; must be paid before common gets anything.
- **Non-cumulative:** skipped dividends **do not** accrue.
Design choice creates the “pressure valve” in stress.

Convertible notes (converts)

Debt that can convert into equity at set terms. Often **low/zero coupon**, pushing cash cost to maturity (refi risk later).

Secured vs. unsecured debt

- **Secured (BTC-collateralized):** has **LTV triggers** → potential **forced BTC sales** if prices fall.
- **Unsecured:** no LTV; selling BTC becomes a management decision, not an automatic event.

SPAC (Special Purpose Acquisition Company)

A blank-check company that raises capital through an IPO to acquire or merge with an operating business, taking it public. In DAT contexts, SPACs are used to launch treasury vehicles but face SEC approval delays (e.g., ~6 months for deSPAC) and restrictions like a 12-month wait for ATM programs due to S-3 eligibility rules. Risks include reclassification as investment companies if lacking genuine operations.

RTO (Reverse Take-Over)

A process where a private company acquires a public shell (often an impaired firm) to gain listing without a traditional IPO. Common for 2025 DAT launches via non-crypto shells (e.g., pharma), RTOs enable quicker capital access but risk legacy liabilities and exclusion from indices like Russell if not evolving into operating businesses.

Liquidity runway

Months the firm can fund **opex + interest + preferred dividends** using cash/available liquidity **without** new issuance or selling BTC.

The flywheel (why premiums matter)

High MNAV → accretive issuance → more BTC/share → story momentum → keeps MNAV high. Works **in bull phases**; reverses fast when MNAV compresses.

Contact

info@marketvector.com

Martin Leinweber

martinleinweber@marketvector.com

Martin Leinweber works as the Digital Asset Product Strategist at MarketVector providing thought leadership in an emerging asset class. His role encompasses product development, research, and communication with the client base of MarketVector. Before joining MarketVector, he worked as a Portfolio Manager for equities, fixed-income, and alternative investments for almost 20 years. Martin was responsible for the management of active funds for institutional investors such as insurance companies, pension funds, and sovereign wealth funds at the leading German quantitative asset manager Quoniam. Previously, he held various positions at one of Germany's largest asset managers, MEAG, the asset manager of Munich Re and ERGO. Among other things, he contributed his expertise and international experience to the establishment of a joint venture with the largest Chinese insurance company PICC in Shanghai and Beijing. Martin is co-author of "Asset-Allokation mit Kryptoassets. Das Handbuch" (Wiley Finance, 2021). It's the first handbook about integrating digital assets into traditional portfolios. He has a Master of Economics from the University of Hohenheim and is a CFA Charter holder.

DISCLAIMER

Copyright © 2025 by MarketVector Indexes GmbH ('MarketVector') All rights reserved. The MarketVector family of indexes (MarketVector™, Bluestar®, MVIS®) is protected through various intellectual property rights and unfair competition and misappropriation laws. MVIS® is a registered trademark of Van Eck Associates Corporation that has been licensed to MarketVector. MarketVector™ and MarketVector Indexes™ are pending trademarks of Van Eck Associates Corporation. BlueStar®, BlueStar Indexes®, BIGI®, and BIGITech® are trademarks of MarketVector Indexes GmbH.

Redistribution, reproduction, and/or photocopying in whole or in part are prohibited without written permission. All information provided by MarketVector is impersonal and not tailored to the needs of any person, entity, or group of persons. MarketVector receives compensation in connection with licensing its indexes to third parties. You require a license from MarketVector to launch any product that is linked to a MarketVector™ Index to use the index data for any business purpose and all use of the MarketVector™ name or name of the MarketVector™ Index. The past performance of an index is not a guarantee of future results.

It is not possible to invest directly in an index. Exposure to an asset class represented by an index is available through investable instruments based on that index. MarketVector does not sponsor, endorse, sell, promote, or manage any investment fund or other investment vehicle that is offered by third parties and that seeks to provide an investment return based on the performance of any index. MarketVector makes no assurance that investment products based on the index will accurately track index performance or provide positive investment returns. MarketVector is not an investment advisor, and it makes no representation regarding the advisability of investing in any such investment fund or other investment vehicle. A decision to invest in any such investment fund or other investment vehicle should not be made in reliance on any of the statements set forth in this document.

Investments into cryptocurrencies and/or digital assets are subject to material and high risk including the risk of total loss. The calculated prices may not be achieved by investors as the calculated price is based on prices from different trading platforms. Furthermore, an investment into cryptocurrencies and/or digital assets may become illiquid depending on the trading platform or investment product used for the specific investment. Investors should carefully review all risk factors disclosed by the relevant trading platform or in the product documents of relevant investment products.

Prospective investors are advised to make an investment in any such fund or other vehicle only after carefully considering the risks associated with investing in such funds, as detailed in an offering memorandum or similar document that is prepared by or on behalf of the issuer of the investment fund or other vehicle. The inclusion of a security within an index is not a recommendation by MarketVector to buy, sell, or hold such security, nor is it considered to be investment advice.

All information shown prior to the index launch date is simulated performance data created from backtesting ("Simulated past performance"). Simulated past performance is not actual but hypothetical performance based on the same or fundamentally the same methodology that was in effect when the index was launched. Simulated past performance may materially differ from the actual performance. Actual or simulated past performance is no guarantee for future results.

These materials have been prepared solely for informational purposes based upon information generally available to the public from sources believed to be reliable. No content contained in these materials (including index data, ratings, credit-related analyses and data, model, software, or other application or output therefrom) or any part thereof (Content) may be modified, reverse-engineered, reproduced or distributed in any form by any means, or stored in a database or retrieval system, without the prior written permission of MarketVector. The Content shall not be used for any unlawful or unauthorized purposes. MarketVector and its third-party data providers and licensors (collectively "MarketVector Parties") do not guarantee the accuracy, completeness, timeliness, or availability of the Content. MarketVector Parties are not responsible for any errors or omissions, regardless of the cause, for the results obtained from the use of the Content. THE CONTENT IS PROVIDED ON AN "AS IS" BASIS. MARKETVECTOR PARTIES DISCLAIM ANY AND ALL EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE, FREEDOM FROM BUGS, SOFTWARE ERRORS, OR DEFECTS, THAT THE CONTENT'S FUNCTIONING WILL BE UNINTERRUPTED OR THAT THE CONTENT WILL OPERATE WITH ANY SOFTWARE OR HARDWARE CONFIGURATION. In no event shall MarketVector Parties be liable to any party for any direct, indirect, incidental, exemplary, compensatory, punitive, special, or consequential damages, costs, expenses, legal fees, or losses (including, without limitation, lost income or lost profits and opportunity costs) in connection with any use of the Content even if advised of the possibility of such damages.